DECK 15 · WEEK 4 · MORNING 3

Duration, Convexity & Interest Rate Risk

The heaviest math block of the course — duration measures, convexity, and curve-based risk for option-free and callable bonds

CFA Level I · Vol 6 Learning Modules 10–13

Roadmap — Fixed Income III: Duration, Convexity & Interest Rate Risk

4 sections · ~2 hours · Built bottom-up from CFA readings

01 **Interest rate risk & horizon yield**

three sources of return · reinvestment vs price risk · BRWA VFO / Baywhite examples · duration gap

02 **Duration: Macaulay, modified, money, PVBP**

MacDur weight tables · ModDur = MacDur/(1+r) · approximation · money duration · PVBP / DV01 · duration drivers

03 **Convexity & the convexity adjustment**

second-order effect · $-MD \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$ · money convexity · portfolio duration & convexity under parallel shifts

04 **Curve-based, effective & key rate**

callables need effective duration · negative convexity · key rate duration for curve shape · empirical vs analytical

Learning objectives

By the end of this deck you should be able to:

1. Decompose the holding-period return on a fixed-rate bond into coupon, reinvestment, and capital-gain components, and explain how reinvestment and price risk offset each other at the Macaulay duration horizon.

2. Compute and interpret Macaulay, modified, money duration and PVBP for a coupon bond, and know when each is the right tool to reach for.

3. Predict how duration moves with coupon, yield, maturity, and time since the last coupon, and rank bonds on interest rate risk without a calculator.

4. Add a convexity adjustment to a duration-based price estimate and quantify, in both percent and currency terms, how much accuracy that second-order term buys.

5. Calculate portfolio duration and convexity as market-value-weighted averages, and state the key assumption (parallel yield-curve shift) and its limits.

6. Decide between yield duration, effective duration, key rate duration, and empirical duration for option-free bonds, callables / MBS, curve-shape views, and credit-risky portfolios.

SECTION 01

Interest rate risk & horizon yield

A fixed-rate bond's total return decomposes into three building blocks: coupon income, reinvestment income, and capital gain or loss on sale

Horizon yield = the compounded annual rate that equates what you pay at t=0 to the total you end up with at horizon T (reinvested coupons + sale price).

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r_horizon  =  [ (FV_coupons + Sale_price) / Purchase_price ]^(1/T)  –  1
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where: FV_coupons = future value of reinvested coupons; Sale_price = price at horizon (or par at maturity); T = years held.

1 **Coupon & principal payments**
The cash itself — promised on scheduled dates. Drops out of interest rate risk as long as the issuer does not default.

2 **Reinvestment of coupons**
Coupons earn the prevailing reinvestment rate until horizon. Rising rates help this leg; falling rates hurt it.

3 **Capital gain / loss on sale**
Only if the bond is sold before maturity. Rising rates push the price down (loss); falling rates push it up (gain) — the opposite of reinvestment.

Source: CFA Institute, Curriculum Vol 6, Learning Module 10, §2 "Sources of Return from Investing in a Fixed-Rate Bond", pp. 239–244.

Holding a bond to maturity strips out price risk but doubles down on reinvestment risk — the buy-and-hold investor wins from rate rises, loses from rate falls

VFO buys BRWA 10-year 6.2% annual-coupon bond at par. Bond held to maturity → no capital gain / loss. Only reinvestment rate moves.

Scenario (rates shift immediately after purchase)	FV of coupons	Redemption	Horizon yield
Rates stable at 6.20% (base case)	82.493	100.000	6.20%
Rates rise +100 bp → 7.20%	86.475	100.000	6.43%
Rates fall −100 bp → 5.20%	78.715	100.000	5.98%

WORKED — VFO HORIZON YIELD IF RATES RISE TO 7.20%

10-year 6.2% annual bond, purchase price 100, coupons reinvested at 7.2% for 10 years, redemption at par.

1. FV of coupons: =FV(7.2%, 10, 6.2, 0) = 86.475 per 100 par
2. Total ending cash: 86.475 + 100 = 186.475
3. Horizon yield: (186.475 / 100)^(1/10) − 1 = 6.43%

Horizon yield = 6.43% (+23 bp vs base case — the gain is pure reinvestment benefit).

[1] Total coupons received = 62.0 (10 × 6.2). In the base case the FV is 82.493 — the extra 20.493 is pure interest-on-interest.

Source: CFA Institute, Curriculum Vol 6, Learning Module 10, Examples 1 and 3 plus the Knowledge Check (VFO), pp. 239–242.

Selling before maturity exposes the investor to price risk that can swamp the reinvestment gain — Baywhite sells at year 4 and ends up losing on a rate rise

Same BRWA 10-year 6.2% bond, but Baywhite has a 4-year horizon. On sale there are 6 years left to maturity.

Scenario (rate moves immediately after purchase)	FV of coupons (4y)	Sale price at t=4	Horizon yield
Rates stable at 6.20%	27.203	100.000	6.20%
Rates rise +100 bp → 7.20%	27.609	95.263	5.28%
Rates fall −100 bp → 5.20%	26.802	105.043	7.16%

WORKED — BAYWHITE WITH RATES UP 100 BP

Coupons reinvested at 7.2% for 4 years (FV = 27.609). Bond sold at t = 4 with 6 years to go, priced off 7.2% YTM.

1. Sale price: =PV(7.2%, 6, 6.2, 100) = 95.263 per 100 par (capital loss 4.737)
2. Total ending cash: 27.609 + 95.263 = 122.872
3. Horizon yield: (122.872 / 100)^(1/4) − 1 = 5.28%

Horizon yield 5.28% — reinvestment gain of 0.406 is swamped by the 4.737 capital loss.

Source: CFA Institute, Curriculum Vol 6, Learning Module 10, Examples 2 and 4 plus the Knowledge Check (Baywhite), pp. 240–243.

The duration gap — Macaulay duration minus horizon — tells you which side of interest rate risk dominates for a given investor

At horizon = MacDur the two offset exactly. Shorter horizon: price risk dominates. Longer horizon: reinvestment risk dominates.

$$\text{Duration gap} = \text{Macaulay duration} - \text{Investment horizon}$$

where: Gap > 0 → price risk dominates (rising rates hurt). Gap < 0 → reinvestment risk dominates. Gap = 0 → nearly hedged.

Investor	Horizon	MacDur (bond)	Duration gap	Dominant risk
VFO (buy-and-hold)	10y	7.74y	−2.26 (neg.)	Falling rates (reinvestment)
Hightest (8y horizon)	8y	7.74y	−0.26 (≈ 0)	Largely hedged
Baywhite (4y horizon)	4y	7.74y	+3.74 (pos.)	Rising rates (price)

IMMUNISATION TAKEAWAY

Matching the Macaulay duration of a bond (or portfolio) to the investor's horizon produces a single-period immunised position against a one-time parallel shift in the yield curve.

[1] Hightest's realised 8y horizon yield: 6.20% stable, 6.24% at +100 bp, 6.17% at −100 bp — near-identical because horizon ≈ MacDur 7.74y.

Source: CFA Institute, Curriculum Vol 6, Learning Module 10, §3 "Investment Horizon and Interest Rate Risk", Exhibit 3 and Equation 1, pp. 245–250.

SECTION 02

Duration: Macaulay, modified, money & PVBP

Macaulay duration is the present-value-weighted average time-to-receipt of a bond's cash flows, expressed in years

Weights sum to 1 because the sum of the PV of cash flows equals the full bond price. MacDur is in units of periods; divide by periods per year to annualise.

$$\text{MacDur} = \Sigma \; t \times [\, PV(CF\_t) \,/\, PV\_full \,]$$

$$\text{Annualised MacDur} = \text{MacDur} \,/\, m$$

where: t = time to each cash flow, in periods; PV_full = full (dirty) price; m = coupon periods per year.

ZERO-COUPON BONDS

Only one cash flow at maturity with weight 1.0. Macaulay duration = time-to-maturity exactly.

A 5-year zero has MacDur = 5y regardless of yield.

PERPETUAL BONDS

No maturity, just infinite coupons. Closed form: MacDur = (1+r) / r.

At r = 6%, a perpetual has MacDur = 1.06 / 0.06 = 17.67y — longest duration of any plain-vanilla fixed income.

Source: CFA Institute, Curriculum Vol 6, Learning Module 10, §4 "Macaulay Duration", Equation 2, pp. 251–255; zero-coupon and perpetual duration, Learning Module 11 §3, pp. 279–280.

Macaulay duration worked example — the 5-year 3.2% semi-annual BRWA bond has an annualised Macaulay duration of 4.66 years

Bond priced at par so weight of each CF = its PV directly. Ten semi-annual periods of 1.6 coupon + 100 face at t=10. Periodic YTM = 1.6%.

Period t	Cash flow	PV @ 1.6%	Weight	t × Weight
1–9 (coupons)	1.6 each	1.57 … 1.39	0.0157 … 0.0139	0.016 … 0.125
10 (coupon + par)	101.6	86.69	0.8669	8.669
Totals	–	100.000	1.0000	9.320

COMPUTE MACDUR (IN PERIODS, THEN ANNUALISED)

Sum of (time × weight) across all 10 periods = 9.3203 semi-annual periods. Periods per year m = 2.

1. Semi-annual MacDur = 9.3203 periods
2. Annualised MacDur = 9.3203 / 2 = 4.6601 years
3. Sanity check: MacDur < 5y because 13.3% of the price sits in coupons paid before maturity

Annualised Macaulay duration = 4.66 years (9.3203 semi-annual periods ÷ 2).

[1] Coupon effect, same bond otherwise: a 5y par bond with a 6.0% semi-annual coupon has MacDur 4.393y. Higher coupon, shorter duration.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, Example 1 "Modified Duration for the BRWA Bond", pp. 268–269.

Modified duration is Macaulay divided by (1 + r) and measures percentage price change per unit yield change — the first-order sensitivity

Duration is quoted as a POSITIVE number; the minus sign lives in the price-change formula. Linear (tangent) approximation — accurate for small yield moves, off for large ones.

$$\text{ModDur} = \text{MacDur} / (1 + r)$$
$$\%\Delta PV\_full \approx -\text{AnnModDur} \times \Delta y$$

where: r = yield PER PERIOD (0.032/2 = 1.6% here); annualise by dividing by m. The error grows with $(\Delta y)^2$, not with Δy.

WORKED — BRWA 5y 3.2% BOND AT 3.2% YTM

MacDur (periods) = 9.3203. Periodic YTM r = 0.016. Annualised modified duration sought.

1. ModDur (periods) = 9.3203 / (1 + 0.016) = 9.17351
2. AnnModDur = 9.17351 / 2 = 4.58676
3. Estimated %ΔPV_full for +80 bp YTM rise: −4.58676 × 0.0080 = −3.67%
4. Estimated %ΔPV_full for −80 bp YTM fall: −4.58676 × (−0.0080) = +3.67%

AnnModDur = 4.587. A 100 bp YTM rise implies ≈ -4.59% price (before convexity).

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §2 "Modified Duration", Equations 2–3, pp. 266–270.

Approximate modified duration bumps the yield up and down by a few basis points and reads the slope directly off the price curve

The preferred method when MacDur is hard to obtain (contingent cash flows, default risk). Yields results essentially identical to the exact computation for small bumps.

$$\text{AnnModDur} \approx (PV_- - PV_+) / [\, 2 \times \Delta\text{yield} \times PV\_0 \,]$$

where: PV_0 = current full price; PV+ / PV− = full prices after raising / lowering YTM by Δyield (annualised).

WORKED — BRWA 5y 3.2% BOND, 5 BP BUMP

PV_0 = 100.00 (priced at par). Δyield = 0.0005 (5 bp). Compute PV+ at 3.25% YTM and PV− at 3.15% YTM.

1. PV+ at 3.25% = 99.770965 per 100 par
2. PV− at 3.15% = 100.229641 per 100 par
3. AnnModDur ≈ (100.229641 − 99.770965) / (2 × 0.0005 × 100.00) = 0.458676 / 0.100000 = 4.58676

Approximate AnnModDur = 4.5868 — matches the exact MacDur/(1+r)/m result to four decimals.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §2 "Approximate Modified Duration", Equation 4 and Example 2, pp. 270–272.

Money duration converts modified duration into a currency figure — the dollar price change per unit yield move on an actual position

US markets call this "dollar duration". Drops the percentage scaling so P&L attribution is in the local currency directly.

$$\text{MoneyDur} = \text{AnnModDur} \times \text{PV\_full}$$
$$\Delta\text{PV\_full} \approx -\text{MoneyDur} \times \Delta y$$

where: PV_full can be in percent-of-par (per 100) or in the actual currency position. Output matches the input unit.

WORKED — USD 100M BRWA 5y BOND POSITION

Par USD 100M. Settlement 57 days into first coupon period → PV_full = 100.504 per 100 par. AnnModDur = 4.43092 (settlement-adjusted).

1. Market value = USD 100,000,000 × 100.504/100 = USD 100,503,921
2. MoneyDur = 4.43092 × USD 100,503,921 = USD 445,324,719
3. Estimated P&L for +100 bp YTM move: −USD 445,324,719 × 0.0100 = −USD 4,453,247
4. Estimated P&L for −100 bp YTM move: +USD 4,453,247 (linear / pre-convexity)

Money duration USD 445.3M. A 100 bp yield rise costs the investor ~USD 4.45M.

[1] MoneyDur inherits the unit of PV_full: 445.32 per 100 of par, or USD 445.3M on the position. Mixing the two is the classic units error.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 "Money Duration and Price Value of a Basis Point", Equations 6–7, p. 278.

PVBP (the "dollar value of an 01") is the currency price change per 1 basis point — the bond-market standard risk number for trading desks

Trading desks typically quote risk in PVBP/DV01 because it scales linearly with notional and lines up with quoted bid-offer in basis points.

$$\text{PVBP} = (\text{PV}- - \text{PV}+) / 2$$
$$\text{where PV}\pm \text{ are full prices at YTM} \mp 1 \text{ bp}$$

where: Per 100 of par or on the actual position notional. PVBP is essentially (MoneyDur × 0.0001) up to convexity rounding.

WORKED — BRWA 5y 3.2% BOND, PVBP PER 100 PAR

PV_0 = 100.504. Bump YTM by ±1 bp (i.e. 3.19% and 3.21%). Use PRICE function for both.

1. PV+ at 3.21% YTM = 100.459400 per 100 par
2. PV− at 3.19% YTM = 100.548465 per 100 par
3. PVBP = (100.548465 − 100.459400) / 2 = 0.044532 per 100 of par
4. Scale to a position: USD 30M par × 0.044532 / 100 = USD 13,360 per bp

PVBP = 0.0445 per 100 of par → DV01 ≈ USD 13.4k on a USD 30M par BRWA position.

[1] Also called "PV01" and, in the US, "DV01" (dollar value of 1 bp). BPV (basis-point value) is money duration × 0.0001.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 "Price Value of a Basis Point", Equation 8, pp. 278–279.

Duration's four drivers — coupon, yield, maturity, and position within the coupon period — each pull in a specific, predictable direction

Lower coupon or yield shifts more weight onto distant cash flows, lengthening duration. Longer maturity does the same via the par payment.

Feature	Change	Effect	Intuition
Coupon rate c	c ↑	Duration ↓	Early cash flows pull the average time in
Yield-to-maturity r	r ↑	Duration ↓	Distant flows deflate more; weight moves near
Time-to-maturity T	T ↑	Duration ↑ *	The par payment sits further out
Time elapsed, t/T	t/T ↑	Duration ↓	Each day, the next coupon is one day closer

RANKING RULE OF THUMB — AND THE ONE EXCEPTION (*)

For two otherwise identical bonds: lower coupon → higher duration; lower yield → higher duration; longer maturity → higher duration. The exception: for a DEEP-DISCOUNT bond (c well below r) at very long maturity, MacDur passes (1+r)/r, peaks, then FALLS — so a 100-year discount bond can carry less rate risk than a 30-year one.

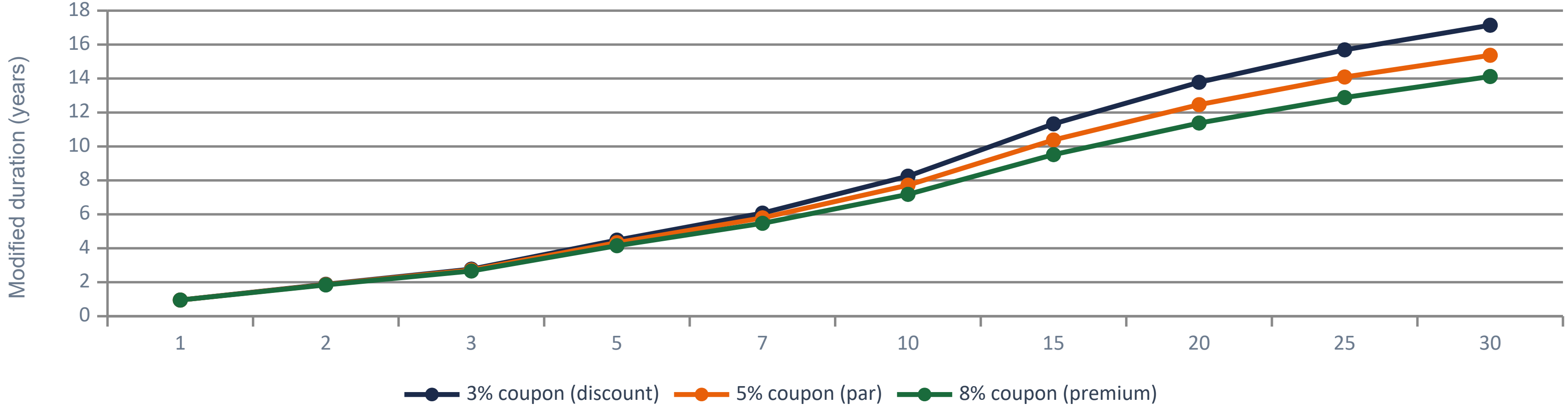
[1] Duration is not static: it falls smoothly through the coupon period, then jumps up on the coupon date — the "saw-tooth" of LM 11 Exhibit 6, p. 286.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §4 "Properties of Duration", Exhibit 4, pp. 284–286.

Duration rises with maturity, falls with coupon, and falls with yield — visualised on one chart across 30 maturities

All three curves share a 5% YTM; the par bond (c = r = 5%) sits between the other two. Plotted in MODIFIED duration, so the long-maturity ceiling is 1/r = 20y, not the Macaulay ceiling (1+r)/r = 21y.

Modified duration vs maturity at 5% YTM — three coupon levels



READING THE CHART

At every maturity the lower-coupon curve sits above — that is the coupon effect. All three flatten toward the perpetuity ceiling, 1/r = 20 years.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, Exhibit 5 "Properties of Macaulay Duration", p. 285. Curves computed from the closed-form Macaulay equation, LM 10 Equation 3, p. 254.

Cross-section stress test: three bonds, same currency, different durations — the expected loss on a 100 bp move scales by more than 20×

Chowdhury holds USD 30M par in each. Same 100 bp yield move — radically different dollar P&L because durations differ by order of magnitude.

Metric	BRWA 5y 3.2%	Australian 1y 0%	Romanian 30y 4.625%
Full price (per 100 par)	100.815	99.262	107.429
Annualised modified duration	4.335	0.737	16.248
PVBP (per 100 par)	0.044	0.007	0.175
Market value (USD 30M par)	30,244,381	29,778,597	32,228,627
Money duration (per 100 par)	437.054	73.163	1745.518
Expected loss, 100 bp YTM rise	−USD 1,311,163	−USD 219,490	−USD 5,236,553

WHY THE ROMANIAN LOSS IS 24× THE AUSTRALIAN

The three positions differ in market value by only ~8%. The loss ratio is 24× because it is driven almost entirely by modified duration (16.25 vs 0.74). Expecting rates to rise, underweight the Romanian; expecting them to fall, overweight it.

[1] Loss = MoneyDur × Δy × (par / 100): 437.054 × 0.01 × 300,000 = USD 1,311,163. The per-100 scaling is where this calculation usually breaks.

Source: CFA Institute, Curriculum Vol 6, Learning Module 11, §3 Question Set (Chowdhury), pp. 281–284.

SECTION 03

Convexity & the convexity adjustment

Duration is a linear approximation; real bond prices curve away from the tangent line — convexity measures that second-order effect

For small Δy, duration alone is fine. For large Δy, the tangent line under-estimates the gain when rates fall and over-estimates the loss when rates rise — convexity corrects both.

DURATION (FIRST-ORDER)

Slope of the price-yield curve at the current YTM. Straight line. Works well for small Δy.

For Δy = 25 bp on a 5y bond, the error is negligible. For Δy = 100 bp on a 30y bond it's material.

CONVEXITY (SECOND-ORDER)

Curvature of the price-yield curve. Always positive for an option-free bond.

Investors like convexity: for a given duration, more convexity = bigger upside from rate falls and smaller downside from rate rises. A free lunch in theory — so priced in.

WHICH WAY DOES DURATION GET IT WRONG? ALWAYS THE SAME WAY

Rates FALL — the true price sits ABOVE the tangent line, so duration UNDER-states the gain. Rates RISE — the true price sits ABOVE the tangent line again, so duration OVER-states the loss. Both errors favour the holder of an option-free bond, and both are corrected by the SAME positive term, $\frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$.

[1] Same price, maturity, YTM and duration: the more convex bond wins in BOTH directions — which is why convexity is priced, not free (Exhibit 3, p. 300).

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §2 "Bond Convexity and Convexity Adjustment", Exhibits 1 and 3, pp. 297–300.

The convexity-adjusted price change formula adds one-half times convexity times the squared yield change to the duration estimate

Second-order Taylor expansion of price(yield). Always makes the estimated price higher (for option-free bonds) vs the linear estimate — regardless of direction.

$$\%\Delta PV\_full \approx -AnnModDur \times \Delta y + \tfrac{1}{2} \times AnnConvexity \times (\Delta y)^2$$

where: Δy = change in annualised YTM, in decimal. First term is duration; the bracket is the convexity adjustment.

$$AnnConvexity \approx [\ PV- + PV+ - 2 \times PV\_0\ ]\ /\ [\ (\Delta y)^2 \times PV\_0\ ]$$

where: Same PV+ / PV– inputs as approximate modified duration — this one ADDS them instead of differencing them.

LOWER COUPON

Pushes weight to the par payment → higher convexity.

LONGER MATURITY

$(\Delta t)^2$ terms dominate → convexity rises faster than duration with T.

LOWER YIELD

Discount factors closer to 1 → weights further out, higher curvature.

[1] Fourth driver — dispersion: of two bonds with the same duration, the one whose cash flows are more spread out has the higher convexity (LM 12, p. 300).

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §2 Equations 1–2 and Exhibit 2, pp. 298–300.

Worked example — BRWA 5y 3.2% bond: duration alone misses the 100 bp price change by 12 bp; the convexity adjustment closes the gap to less than 0.3 bp

AnnModDur = 4.58676. AnnConvexity = 24.23895. Δy = ±0.01. Compare linear, convexity-adjusted, and exact PRICE-function values.

Δy	Exact %ΔPV	ModDur only	Err (bp)	ModDur + ½·Con·Δy²	Err (bp)
+100 bp	−4.46788%	−4.58676%	−11.89	−4.46556%	+0.23
−100 bp	+4.71035%	+4.58676%	−12.36	+4.70795%	−0.24

WALK-THROUGH — Δy = +100 BP

Linear: −4.58676 × 0.01 = −4.587%. Convexity adjustment: ½ × 24.23895 × $(0.01)^2$ = +0.00121 = +12.1 bp.

1. Duration contribution: −4.58676 × 0.0100 = −0.045868 = −4.587%
2. Convexity contribution: ½ × 24.23895 × $(0.0100)^2$ = +0.001212 = +0.121%
3. Total estimate: −4.587% + 0.121% = −4.466%
4. Exact (PRICE function at 4.20% YTM): 95.53212 → −4.468% → error only 0.2 bp

Duration + convexity lands within 0.3 bp of exact; duration alone off by 12 bp in both directions.

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §3 "Bond Risk and Return Using Duration and Convexity", Exhibit 4, pp. 303–304.

Money convexity scales convexity to currency units — the extra dollar adjustment on a position's price change over what money duration predicts

Same second-order logic, now in currency. Adds precision to P&L forecasting on large positions where the convexity term becomes material.

$$\text{MoneyCon} = \text{AnnConvexity} \times \text{PV\_full}$$
$$\Delta\text{PV\_full} \approx -\text{MoneyDur} \times \Delta y + \tfrac{1}{2} \times \text{MoneyCon} \times (\Delta y)^2$$

WORKED — USD 100M BRWA POSITION, Δy = +100 BP

AnnModDur = 4.58676, AnnConvexity = 24.23895, PV_full at par = USD 100,000,000.

1. MoneyDur = 4.58676 × 100,000,000 = USD 458,675,875
2. MoneyCon = 24.23895 × 100,000,000 = USD 2,423,894,503
3. Duration leg: −458,675,875 × 0.01 = −USD 4,586,759
4. Convexity leg: ½ × 2,423,894,503 × $(0.01)^2$ = +USD 121,195
5. Net estimate: −USD 4,465,564. Actual price drop: −USD 4,467,884. Error: −USD 2,320.

Convexity term trims the duration-only over-estimate by ~USD 121k, closing 98% of the error.

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §3 Equations 3–4 and Exhibit 5, pp. 304–305.

Convexity is always positive for an option-free bond and rises with maturity, falls with coupon and yield — the same drivers as duration but amplified

Because the convexity formula involves t × (t+1), the maturity effect scales like T^2 — long bonds have enormously more convexity than short ones.

Bond	Maturity	Coupon	YTM	AnnModDur	AnnConvexity	Con / Dur
BRWA (semi)	5y	3.20%	3.20%	4.59	24.24	5.3
Romanian (annual)	30y	4.625%	4.75%	15.91	369.64	23.2
Gap	×6	–	–	×3.5	×15.3	×4.4

WHAT THIS MEANS FOR A LONG-BOND PORTFOLIO

A 100 bp yield rise on the Romanian bond: duration piece −15.91%, convexity piece +1.85%, net −14.06%. The convexity term is 15× BRWA's — no longer a tiny correction but a material component of the price change. Note that the Romanian carries the HIGHER coupon and the HIGHER yield, both of which shorten duration and cut convexity; its 30-year maturity overwhelms them regardless. For long-duration books, ignoring convexity is a systematic mis-estimation, not a rounding error.

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Example 1 "Duration and Convexity of 30-Year Romanian Bond", pp. 300–301.

Portfolio duration is the market-value-weighted average of constituent durations — straightforward to compute, but assumes parallel yield curve shifts

Weights are market values (not par, not face). Formula below extends identically to convexity. Bonds ought to share the same reporting currency.

$$\text{AnnModDur\_port} = \Sigma_i\ w_i \cdot \text{AnnModDur}_i$$
$$\text{AnnConvexity\_port} = \Sigma_i\ w_i \cdot \text{AnnConvexity}_i$$

where: w_i = MV(bond i) / MV(portfolio); weights sum to 1. Same price-change formula applies with these portfolio-level measures.

WORKED — BRWA + ROMANIAN TWO-BOND PORTFOLIO (USD 50M PAR EACH)

BRWA priced at 100.000 → MV USD 50.00M (w = 0.5050). Romanian priced at 98.022 → MV USD 49.01M (w = 0.4950). Total MV USD 99.01M.

1. Portfolio AnnModDur = 4.58676 × 0.5050 + 15.90637 × 0.4950 = 10.19004
2. Portfolio AnnConvexity = 24.23896 × 0.5050 + 369.64203 × 0.4950 = 195.21581
3. 100 bp parallel UP shift: $\%\Delta PV = -10.190 \times 0.01 + \frac{1}{2} \times 195.216 \times (0.01)^2 = -9.214\%$
4. Equal PAR does not mean equal weight: the weights above are MARKET values, and they differ by 1 pt

Portfolio value falls ~USD 9.12M on a 100 bp parallel rise — provided the shift really is parallel.

[1] The theoretically correct method uses the aggregate portfolio cash flows. MV-weighting is standard practice, and the parallel-shift assumption is baked into it.

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, §4 "Portfolio Duration and Convexity", Exhibit 6, pp. 307–308.

GBP 50M position, 10-year 3.50% bond — investor expects a 100 bp rate decline; the full duration-plus-convexity estimate is +8.78%, not +8.38% from duration alone

For a 100 bp move the convexity term is 40 bp of the total estimate — large enough to matter for position sizing and stop-loss setting.

$$\%\Delta PV\_full = (-8.376 \times -0.01) + \tfrac{1}{2} \times 81.701 \times (-0.01)^2 = 0.08376 + 0.004085 = +8.78\%$$

WORKED — GBP 50M POSITION · ANNMODDUR 8.376 · ANNCONVEXITY 81.701

10y 3.50% semi-annual coupon bond, priced at par. Investor's view: −100 bp parallel rate decline. Compute both the percent price change and the money convexity adjustment.

1. Duration leg: −8.376 × (−0.01) = +0.08376 = +8.376%
2. Convexity leg: ½ × 81.701 × $(-0.01)^2$ = +0.004085 = +0.409%
3. Combined estimate: +8.785%
4. MoneyDur = 8.376 × GBP 50,000,000 = GBP 418,800,000 → duration leg = GBP 4,188,000
5. MoneyCon = 81.701 × GBP 50,000,000 = GBP 4,085,033,604
6. Money convexity adjustment: ½ × 4,085,033,604 × $(0.01)^2$ = +GBP 204,252

GBP 4.188M (duration) + GBP 0.204M (convexity) = GBP 4.39M expected gain on −100 bp.

Source: CFA Institute, Curriculum Vol 6, Learning Module 12, Learning Module Self-Assessment Q3 and Q4, pp. 295–296.

SECTION 04

Curve-based, effective & key rate duration

Yield-based duration assumes cash flows are certain — it breaks down for callable, puttable, or mortgage-backed securities whose cash flows depend on rates

A callable bond has no single well-defined YTM because the issuer can redeem early at the call price if rates fall — so MacDur and ModDur lose meaning. Cue effective duration.

CALLABLE BOND

Issuer holds a call option. When rates FALL far enough, issuer redeems at call price → investor loses upside.

Curve: tangent flattens as yields fall, then curve becomes NEGATIVELY convex. Effective duration < straight-bond duration when rates are low.

PUTABLE BOND

Investor holds the put option. When rates RISE far enough, investor sells back at par → downside truncated.

Curve: flattens on the rising-rate side. Effective duration < straight-bond duration when rates are high. Convexity stays positive.

MBS / ABS

Prepayment option held by homeowners. Falling rates → refinance wave → early principal return → same flattening + negative convexity as callables.

Effective duration from an OAS model is the only usable risk number.

[1] A callable bond has no well-defined YTM — yield-to-worst prices just one of many rate paths, so MacDur and ModDur lose their meaning (LM 13, p. 321).

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §2 "Curve-Based Interest Rate Risk Measures", Exhibits 1–2, pp. 320–322.

Effective duration and effective convexity measure price sensitivity to a parallel shift of the benchmark yield curve — valid whether cash flows are certain or not

Formulas look like the approximate-duration ones — but the bump is on the benchmark curve, not on the bond's own YTM. PV± come from an option-valuation model.

$$\text{EffDur} = (PV_- - PV_+) / [\, 2 \times \Delta\text{Curve} \times PV_0 \,]$$

where: ΔCurve = parallel shift of the benchmark (e.g. government par) curve. PV± come from an option-pricing model.

$$\text{EffCon} = [\, PV_- + PV_+ - 2 \times PV_0 \,] / [\, (\Delta\text{Curve})^2 \times PV_0 \,]$$

where: Can be negative for callables / MBS. $\%\Delta PV_{full} \approx -\text{EffDur} \times \Delta\text{Curve} + \frac{1}{2} \times \text{EffCon} \times (\Delta\text{Curve})^2$.

WHY MODDUR ≠ EFFDUR (EVEN FOR OPTION-FREE BONDS)

A parallel shift in the government PAR curve does not produce a parallel shift in the SPOT curve, so the bond's own YTM moves by a slightly different amount than the par-curve bump. EffDur and ModDur therefore differ a little even for a plain bond. The gap narrows when the curve is flatter, the maturity shorter and the bond closer to par — and disappears only for a flat curve.

[1] BRWA 5y, ±5 bp: a YTM bump gives ModDur 4.587 / Con 24.24; the same bump to the par CURVE gives EffDur 4.816 / EffCon 40.0 (LM 13, pp. 323–324).

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §2 Equations 1–2 and §3 Equation 3, pp. 322–327.

Callable bonds can have negative effective convexity when rates fall below the coupon — the issuer's call option truncates the upside

PM targets duration between 7 and 8 with positive convexity. The bond passes the duration screen but fails on convexity — recommend not to buy.

Scenario	Curve move	Full price	From the model
Base case (today)	0	101.060	PV_0 — observed price
Benchmark par curve +25 bp	+0.0025	99.050	PV+ — option model
Benchmark par curve −25 bp	-0.0025	102.891	PV- — option model

EFFECTIVE DURATION AND CONVEXITY

Plug PV_0 = 101.060, PV+ = 99.050, PV− = 102.891, ΔCurve = 0.0025 into the formulas.

1. EffDur = (102.891 − 99.050) / (2 × 0.0025 × 101.060) = 3.841 / 0.5053 = 7.601
2. EffCon = (102.891 + 99.050 − 202.120) / [$(0.0025)^2$ × 101.060] = −0.179 / 0.00063163 = −283.4
3. Duration clears the 7–8 screen; negative convexity means a rally delivers LESS than duration suggests

EffDur 7.60, EffCon -283. At ±100 bp: -9.03% down vs only +6.17% up. Decline.

[1] Example 2, p. 328, at ±100 bp on the same bond: −7.601 × 0.01 + ½(−285.168)$(0.01)^2$ = −9.03%, while the same shift down gives only +6.17%.

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, Example 1 "Effective Duration and Convexity", p. 324; the ±100 bp asymmetry is Example 2, p. 328.

Key rate duration (partial duration) is the bond's sensitivity to a change in one point on the benchmark curve — how the bond responds to a steepening or twist

EffDur compresses the whole curve into one number. KRD breaks it apart by tenor, letting a manager diagnose "shaping risk" and tilt against a view of the curve shape.

$$\text{KeyRateDur\_k} = -(1 / PV) \times (\partial PV / \partial r\_k)$$
$$\Sigma_k \text{ KeyRateDur\_k} = \text{EffDur}$$

where: r_k = benchmark yield at the kth key maturity (2y, 5y, 10y, 30y...). Summed across the curve, key rate durations give back effective duration.

Tenor	2y bond	5y bond	10y bond	Portfolio (MV wts)
Key rate 2y	1.990	–	–	0.711
Key rate 5y	–	4.938	–	1.675
Key rate 10y	–	–	9.828	2.981
Sum = Effective duration	1.990	4.938	9.828	5.368

USE CASE

Expect the long end to sell off? Cut 10y key rate duration and add at the front. A single portfolio duration cannot express that trade.

[1] USD 100M par of each zero → MV 99.0 / 94.0 / 84.0M, weights 35.7 / 33.9 / 30.3%. Portfolio KRD at a tenor = that bond's MV weight × its own duration.

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §4 "Key Rate Duration as a Measure of Yield Curve Risk", Equations 4–6 and Exhibit 5, pp. 329–331.

Key rate duration worked example — under a bear steepening BOTH bonds lose, but the 10-year loses roughly twice as much

Bear steepener: 1y up 100 bp, 5y up 150 bp, 10y up 200 bp, 20y up 250 bp, 30y up 300 bp. Equal positions in two bonds — what do you do?

Bond	Key tenor	KeyRateDur	Shift at tenor	Estimated %ΔPV
Bond A	5y	2.702	+150 bp	$-2.702 \times 0.015 = -4.05\%$
Bond B	10y	3.953	+200 bp	$-3.953 \times 0.020 = -7.91\%$

DECISION — WHAT THE KRDs ACTUALLY TELL YOU

Multiply each bond's key rate duration by the forecast shift AT ITS OWN TENOR — not by a single parallel number.

1. Bond A: $-2.702 \times 0.0150 = -4.05\%$. Bond B: $-3.953 \times 0.0200 = -7.91\%$
2. CFA answer: sell BOTH — every tenor is forecast higher, so both positions lose
3. If only one can go, it is Bond B: the bigger KRD is hit by the bigger shift, ≈ 2× the loss

Both bonds lose. Sell both; if forced to choose one, Bond B at -7.91%.

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §4 Question Set, Questions 1–3, pp. 332–333.

Empirical duration replaces mathematical formulas with historical data; it's the right measure for credit-risky bonds in market stress, where yields and spreads move in opposite directions

Analytical (MacDur, ModDur, EffDur, KRD) assumes yields and spreads are independent. Empirical regression picks up the real-world correlation — usually negative in flight-to-quality episodes.

ANALYTICAL DURATION

Formula-based from a bond's own cash flows or an OAS model. Assumes spreads are uncorrelated with the benchmark.

Works well for: governments, high-IG corporates, benign regimes.

Fails when: credit spreads widen as benchmarks rally — late Feb to Mar 2020, the COVID flight-to-quality episode.

EMPIRICAL DURATION

Regression of HISTORICAL bond returns on benchmark yield changes. Captures the spread-yield correlation directly.

For HY bonds during market stress, empirical duration is MATERIALLY LOWER than analytical — because widening spreads partly offset falling benchmarks on the bond price.

Challenges: factor choice, sample window, regime stability.

HOW TO CHOOSE — IT IS A CORRELATION QUESTION, NOT A MATHS ONE

If benchmark yields and the bond's spread are effectively uncorrelated — governments, high-grade credit, calm markets — analytical duration is fine. Where they move against each other — high yield, stressed markets — empirical duration is the more accurate forecast of the price change.

Source: CFA Institute, Curriculum Vol 6, Learning Module 13, §5 "Empirical Duration", Exhibits 6–7 and Example 4, pp. 333–336.

KEY TAKEAWAYS

Deck 15 — Fixed Income III: Duration, Convexity & Interest Rate Risk

- Total return = coupons + reinvestment income + capital gain/loss on sale. Reinvestment and price risk move in OPPOSITE directions with rates.
- Duration gap = MacDur − horizon. Positive → price risk, fear rising rates. Negative → reinvestment risk, fear falling rates. Zero → nearly immunised.
- The chain: MacDur in PERIODS → ÷(1+r per period) → ÷m → AnnModDur (% per Δy) → ×PV_full → MoneyDur (currency) → ×0.0001 → PVBP (currency per bp).
- Duration is quoted POSITIVE; the minus lives in the formula: %ΔPV ≈ −ModDur × Δy. Lower coupon, lower yield, longer maturity → higher duration.
- Convexity adjustment $\%\Delta PV \approx -MD \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$. On BRWA at 100 bp it cuts a 12 bp duration-only error to under 0.3 bp — and it is positive both ways.
- Portfolio duration and convexity are MARKET-VALUE-weighted averages. Fast, standard, and silently assuming a PARALLEL shift — which masks steepening risk.
- Bonds with embedded options need EFFECTIVE duration: bump the benchmark CURVE, reprice with an option model. Callables can go negatively convex (EffDur 7.60, EffCon −283).
- Key rate durations split exposure by tenor and sum back to EffDur. Empirical duration beats analytical for credit-risky bonds in stress, where spreads and benchmark yields move opposite ways.